DEBRE BIREHAN UNIVERSITY
COLLAGE OF COMPUTING AND INFORMATICS
DEPARTMENT OF INFORMATION TECHNOLOGY
PROJECT PROPOSAL TITLE: MAINTENANCE WORKFLOW AND TRACKING SYSTEM

PREPARED BY:

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1.Introduction

1.1. Background of the Project

Information and Communication Technology (ICT) play a critical role in modern educational institutions. Universities rely heavily on computers, networks, printers, and other IT equipment for teaching, learning, and administrative tasks. When these devices malfunction, a quick and efficient maintenance response is essential. Traditionally, requesting IT maintenance involves physically visiting the office and filling out a paper form. While this manual method has been used for years, it is prone to inefficiencies, misplacement of records, and delays in service delivery. This project aims to digitize the maintenance process at Debre Birhan University by developing a web-based request and tracking system.

2. Statement of the Problem

Currently, Debre Berhan University utilizes a manual, paper-based maintenance workflow and request system. Staff and students face several challenges with this existing system:

Time-consuming: Users must leave their workplace or dormitories to physically submit a paper form at the ICT office.
Lack of Tracking: Once a request is submitted, users cannot track the status of their maintenance request.
Data Loss and Poor Record Keeping: Paper forms can easily be lost or damaged, making it difficult for the ICT department to keep an accurate history of repaired equipment.
Inefficient Dispatching: IT technicians lack a centralized dashboard to prioritize urgent issues and manage their daily tasks efficiently.

3. Objectives of the Project

3.1. General Objective

The main objective of this project is to develop and implement a Maintenance workflow and Tracking System for Debre Birhan University to streamline the workflow between users and the ICT department.

3.2. Specific Objectives

To achieve the general objective, the following specific objectives are set:
To replace the manual, paper-based maintenance request form with a digital web form.
To allow users to submit IT issues remotely from anywhere within the campus.
To provide a real-time status tracking feature for users to monitor their requests.
To create an admin dashboard for ICT technicians to receive, manage, and update maintenance tickets.

4. Scope of the Project

The scope of this project is limited to managing maintenance requests within Debre Berhan University. The system will handle hardware and network failure reports submitted by the university's academic staff, administrative workers, and students. The platform will be accessible via standard web browsers. However, the system will not cover external maintenance tracking outside the university, nor will it include financial billing or inventory purchasing processes.

5. Significance of the Project

The implementation of this system will bring significant benefits to Debre Birhan University:

***For Users:** It provides a convenient and fast way to report issues without leaving their offices.*

***For Technicians:** It offers a clear, prioritized list of daily tasks, improving their response time and efficiency.*

***For the University Administration:** It eliminates paper waste, ensures accountability, and provides digital records to help management make informed decisions about replacing frequently broken equipment.*

Priority Levels

1. High Priority:** **This level applies to critical failures that cause a complete outage of campus-wide or department-wide core services. Examples include main server downtime, core network switch failures, or administrative office network outages during examination periods. For this priority level, the technician response time is less than 30 minutes, with a target resolution time within 4 hours.

2. Medium priority:** **This covers critical hardware or software failures that significantly affect multiple users or key administrative staff. Examples include classroom projector failures right before lectures, computer lab printer outages, or key departmental staff PC malfunctions. For this level, the technician response time is less than 2 hours, and the target resolution time is within 24 hours.

3. Low Priority: ***This level handles individual, non-critical hardware issues, minor software problems, or non-urgent technical bugs. Examples include replacing an individual staff member's mouse or keyboard, routine software installations, or minor display glitches. For this level, the response time is less than 4 hours, and the issue should be resolved within 48 hours.***

6. Technical Stack & Architecture

- ***Frontend:*** *HTML, CSS and JavaScript.*
- ***Backend:*** *PHP, Python and Node.js.*
- ***Database:*** *MySQL and PostgreSQL.*
- *Version Control: Git, GitHub*
- *Deployment: Netlify, Render, MongoDB Atlas*
- *AI Support: ChatGPT, GitHub Copilot, Claude*
- *All selected technologies are: free, widely used, and suitable for academic projects.*

7.Project Timeline (August 1 – August 30)

The implementation plan spans exactly 30 days, structured to cover the entire development lifecycle from initial design to final deployment. The first week focuses on database structuring, system requirements, and core module implementation. The mid-period (August 8–25) is dedicated to full dashboard logic, API integration, asset management, and rigorous security testing. Finally, the last five days are allocated for bug fixes, system deployment on DBU servers, and final documentation.

Time frame	*Project Phase*	*Activities*
August 1 – August 3	***Requirements Analysis & Database Setup***	***• System requirements gathering and workflow mapping.*** ***• Database table setup (MySQL/PostgreSQL) and database structure configuration.***
August 4 – August 7	***Core Module Development***	***• UI/UX wireframing and initial frontend page templates setup.*** ***• User Login and Request Submission form development.***
August 8 – August 13	***Dashboards & Tracking Logic Implementation***	***• Backend logic for ticket routing, assignment, and status updates.*** ***• Frontend implementation of Technician and Admin dashboards.***
August 14 – August 19	***System Integration & Asset Management***	***• Connecting Frontend UI components directly with Backend API endpoints.***
August 20 – August 25	***Testing, Security Verification & Optimization***	***• Unit Testing, Integration Testing, and User Acceptance Testing (UAT).***
August 26 – August 28	***System Refinement & Bug Fixing***	***• Final system debugging and performance optimization.*** ***• Preparation of server environment for production deployment.***
August 29 – August 30	***Deployment & Final Documentation***	***• Deployment of the system onto the Debre Birhan University local server.***

8.Project Budget Plan

No.	*Budget Category / Description*	*Detail / Unit*	*Rate (ETB)*	*Total Cost (ETB)*
1	*Hardware Equipment*	*Laptops/PCs & Test Mobile Devices (For testing UI responsiveness & API)*	*Lump sum*	*50,000 ETB*
2	*Development & Technical Labor*	*3 Developers / 1 Month*	*6,333.33 / person*	*19,000 ETB*
3	*Transport & Mobility Allowance*	*3 Members / 4 Weeks (Local Travel for Requirements & Testing)*	*1,000 / person*	*3,000 ETB*
4	*Connectivity & Communication*	*Data & Internet Services (4 Weeks)*	*1,000 / package*	*3,000 ETB*
5	*Software Stack & Hosting Setup*	*Open Source Stack & Local DBU Server Setup*	*Lump sum*	*4,000 ETB*
6	*Documentation & Printing*	*Project Reports & Operational Manuals Printing*	*Lump sum*	*2,000 ETB*
7	*Contingency Fund*	*Emergency & Unexpected Costs (~10%)*	*Reserve*	*8,100 ETB*
TOTAL	*GRAND TOTAL PROJECT BUDGET*			*89,100 ETB*

9.Team Collaboration

Team Member	*Role*	*Week 1 Tasks*	*Week 2 Tasks*	*Week 3 Tasks*	*Week 4 Tasks*
Meseret Ayele	***Backend Developer***	***System Architecture & Authentication Logic Setup (RBAC, Password Hashing)***	***Request Submission & Ticket Routing API Endpoints Development***	***Status Update, Resolution & Report Generation Backend Logic***	***API Integration Testing, Server Logic Optimization & Bug Fixes***
Sosina Tezera	***Frontend Developer***	***UI/UX Wireframing & Layout Structure Design (React/HTML/CSS)***	***User Login Page & Digital Request Form Frontend Implementation***	***Technician Dashboard & Real-Time Ticket Tracking Interfaces***	***UI Responsiveness, Cross-Browser Testing & User Experience Refinement***
Takele Moges	***Database Developer & Analyst***	***Database Schema Design, Entity-Relationship Diagram (ERD) & Requirement Analysis***	***Relational Database Setup (MySQL/PostgreSQL) & Table Normalization***	***Complex SQL Queries, Stored Procedures & Data Analytics Views Creation***	***Database Performance Tuning, Data Security Verification & Backup Config***

10. Testing & Quality Assurance

***Unit Testing**: Verifying form field validations, password security, and status changes.*

***Integration Testing**: Ensuring seamless data flow between Requisitioner -> Technician -> Supervisor Approval.*

***User Acceptance Testing (UAT)**: Testing system usability directly with DBU ICT technicians to ensure complete alignment with original paper procedures.*

***Security Testing**: Enforcing role permission checks and preventing SQL injection and unauthorized data access.*

11. Expected Impact & Benefits

***Elimination of Paper Waste:** Replaces physical paper forms with secure, centralized digital records.*

***Time Savings:** Eliminates physical travel across campus to submit requests.*

***Data-Driven Insights**: Provides ICT management with clear analytics on frequently failing hardware to plan future purchases.*

12. Use Case Diagram

The Use Case Diagram illustrates the main interactions between three primary actors and the system:

***End User (Staff/Student):** Can Login, directly Submit Requests, View Request, Check Status, Add Comments ensuring quick and easy access.*
***Admin:** Must securely Login to Manage Users, Manage Assets, System Config, and View All Data .*
***Technician:** Must securely Login to View Assigned Request, Update Status & Add Resolution (e.g., 'Fixed'), Reopen Request and Close Request*
This design provides a fast, paperless process for users while keeping the IT management secure and well-organized.

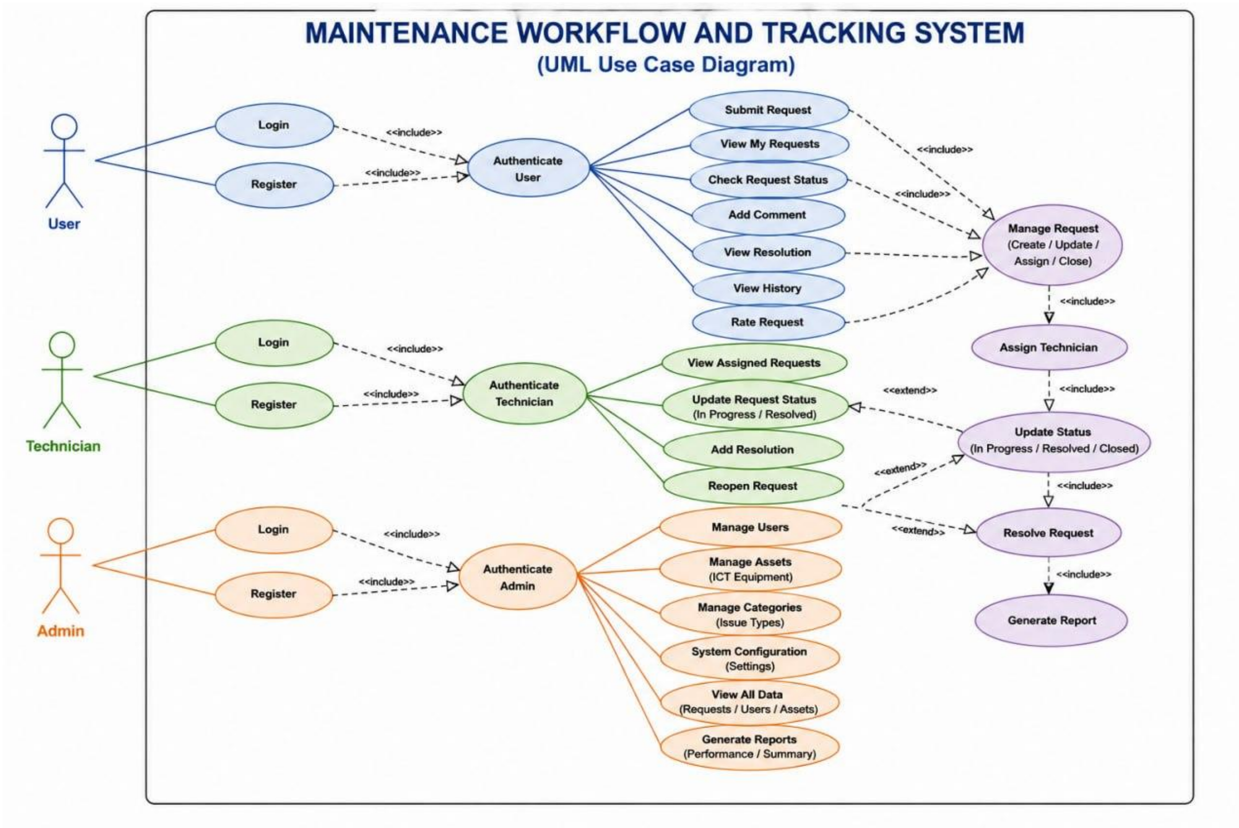

Figure 1: Use Case Diagram of The Maintenance workflow and tracking System

13. Conclusion

The Maintenance workflow and tracking system will transform Debre Birhan University IT support workflow by replacing the outdated paper voucher system with an efficient digital platform. By enabling remote request submissions, real-time ticket tracking, and automated failure reporting, the system significantly reduces service delays, improves operational accountability, and optimizes technician productivity all within a manageable 1-month development timeline.